

Design Rationale for Mini Library Management System

1. Overview

This project implements a simple Library Management System in Python using core data structures — dictionaries, lists, and tuples. It allows adding, updating, deleting, borrowing, and returning books, as well as managing members. The design prioritizes simplicity, readability, and efficiency for small-scale systems or learning environments.

2. Data Structures Used

- **Dictionary (books):** Stores books using ISBN as key for fast lookup and modification.
- **List (members):** Maintains a collection of member records, allowing multiple dictionaries (member profiles).
- **Tuple (genres):** Used for storing a fixed set of valid book genres to ensure immutability and data consistency.

3. Benefits

- Enables fast access and update of records using keys.
- Flexible structure for CRUD operations.
- Easy to extend for larger systems or database integration.
- Enhances readability and maintainability for beginner-level projects.

4. Limitations

- Data is stored in memory (no database persistence).
- Designed for small data volumes.
- No graphical interface.

5. Conclusion

The choice of Python data structures ensures a balanced mix of simplicity and performance. Dictionaries provide quick data retrieval, lists allow ordered member management, and tuples guarantee stable category references. Together, they form an efficient model for a mini library management system.